

(Team Name)

(SMART IOT-BASED PARKING SYSTEM)

Software Design Document

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1.0 INTRODUCTION

1.1 Purpose

The purpose of this Software Design Document (SDD) is to describe the architectural design, system components, database structure, and user interface design of the Smart IoT-Based Parking System. This document serves as a technical reference for developers, designers, testers, and project evaluators by explaining how the system will be implemented to satisfy the requirements specified in the Software Requirements Specification (SRS).

The SDD provides a detailed design of the system modules, data management processes, communication between components, and user interactions. It acts as a blueprint for the development and maintenance of the Smart IoT-Based Parking System.

1.2 Scope

The main goal of this Smart IoT-Based Parking System is to develop a centralized, real-time parking management that helps to improve parking management efficiency through real-time monitoring and IoT technology. The system will use the IoT sensors in monitoring the occupancy of the parking slots and integrate it with a platform of mobile application. The system aims to reduce the bottleneck and utilize the parking space and also provide accurate information to the users up-to-the-minute.

The objective(s) of the system are:

- a) To provide a real-time parking availability information through a mobile application for the drivers
- b) To automate the updates of the parking status by using the IoT sensors technology.
- c) To maintain a centralized database to store the data of the parking, users, sensors and transactions information.
- d) To provide a platform for the administrators to monitor the parking occupancy, manage the parking slots and generate reports.

- e) To establish stable communication between the IoT sensors and the Central Management Platform (CMP).

The system gives benefit(s) like:

- i. Reduce time and fuel wasted on finding an open parking slot by checking the availability via a mobile application.
- ii. Improves the utilization of parking spaces by monitoring in real-time.
- iii. Providing accurate real-time information of the parking availability.
- iv. Enables parking operations to be monitored through a centralized database by the administrators.
- v. Enhances users' convenience by using a smart parking management technology-based system.

1.3 Overview

This document is organized into several sections that describe the design and implementation of the Smart IoT-Based Parking System.

Section 1 introduces the purpose, scope, references, and terminology used throughout this document. Section 2 provides an overview of the system and describes its major functionalities. Section 3 explains the system architecture, including subsystem decomposition and the design rationale behind the proposed system structure.

Section 4 presents the data design, including the database structure and data dictionary used to manage system information. Section 5 describes the detailed design of system components, including their functionalities and processing logic. Section 6 explains the human interface design and user interaction screens to demonstrate how users interact with the system.

Section 7 provides the requirements traceability matrix, which links the design components to the requirements specified in the Software Requirements

Specification (SRS). Finally, Section 8 contains appendices and supporting information related to the system design.

1.4 Reference Material

1. IEEE, IEEE Guide to Software Requirements Specifications, IEEE Std 830-1998, IEEE, New York, NY, USA, 1998. [Online]. Available: [IEEE Software Requirements Specification Template.pdf](#)
2. Espressif Systems, ESP32 Series Datasheet, 2025. Accessed: May 14, 2026. [Online]. Available: [Wireless SoCs, Software, Cloud and AIoT Solutions | Espressif Systems](#)
3. Oracle, MySQL 8.0 Reference Manual, 2025. Accessed: May 14, 2026. [Online]. Available: [MySQL :: MySQL 8.0 Reference Manual](#)
4. PayPal, PayPal Developer Documentation, 2025. Accessed: May 14, 2026. [Online]. Available: [Get Started with PayPal REST APIs](#)
5. Task 6: Smart IoT-Based Parking System, BITP 2223 Software Requirement and Design, Universiti Teknikal Malaysia Melaka, 2026. [Online]. Available: [LAB TASK 6 - SRD.pdf](#)
6. SRS Document: Smart IoT-Based Parking System, BITP 2223 Software Requirement and Design, Universiti Teknikal Malaysia Melaka, 2026. [Online]. Available: [SRS DOCUMENT.pdf](#)

1.5 Definitions and Acronyms

Term	Definition
Admin	Authorized user who monitors parking occupancy, manages slots, and generates reports Task-6-1.pdf.
Availability	Current status of a parking slot, either occupied or available Task-6-1.pdf.
CMP	Central Management Platform, the backend system that processes sensor data and supports administration Task-6-1.pdf.
Driver	End user who logs in, checks parking availability, and makes payments Task-6-1.pdf.
HC-SR04	Ultrasonic sensor used to detect vehicle presence in a parking slot Task-6-1.pdf.
IoT	Internet of Things; the sensor-based connection between parking hardware and the software system Task-6-1.pdf.
LPWAN	Low Power Wide Area Network, used for low-power communication between sensors and the system Task-6-1.pdf.
Payment History	Record of past parking transactions made by a driver Task-6-1.pdf.
Parking Slot	A single parking space monitored by the system Task-6-1.pdf.
Real-Time Data	Data sent immediately from sensors to update slot availability Task-6-1.pdf.
Sensor	Hardware component that detects vehicle presence or absence Task-6-1.pdf.
Slot Management	Administrative function for adding, updating, or deleting parking slot information Task-6-1.pdf.

Transaction	Payment record stored in the database after a parking fee is processed Task-6-1.pdf.
UC	Use Case, a documented user-system interaction flow such as UC-01 to UC-08 Task-6-1.pdf.
Unavailable / Inactive	Slot state shown when the sensor is not responding or data cannot be received Task-6-1.pdf.

2.0 SYSTEM OVERVIEW

The Smart IoT-Based Parking System is a comprehensive web and mobile-based parking management solution developed to enhance parking efficiency utilizing real-time monitoring and IoT technology. The system functions as a centralized platform that seamlessly connects drivers, administrators, and IoT sensor devices to facilitate effective parking management.

At its core, the system integrates IoT sensors installed directly at parking slots with a Central Management Platform (CMP) and a robust database system. These hardware sensors actively detect vehicle presence and instantly transmit real-time parking data to the central system. This architecture relies heavily on continuous internet and Wi-Fi connectivity to ensure communication between the sensors and the central database. All slot statuses, user activities, payment histories, and reporting data are securely stored and continuously updated within a centralized database to ensure highly reliable access.

Functionality and Context

The system is partitioned into three primary functional areas based on user roles and components:

- **Driver Functions:** Drivers utilize the mobile or web application to securely log in, view real-time parking availability (visualized on a map or via highlighted markers), search for spaces, and process parking payments digitally.
- **Administrative Functions:** Administrators have access to a dedicated dashboard to

continuously monitor overall parking occupancy, manually manage parking slots (add, update, or delete), and generate analytical reports regarding parking usage and revenue.

- IoT Sensor Functions: The automated hardware subsystem continuously detects the presence or absence of vehicles in respective slots. It dynamically updates the central database with real-time slot statuses (e.g., occupied, available, or inactive in the event of a failure) without requiring human intervention.

The system follows a three-tier architecture consisting of:

- Presentation Layer: Mobile and web interfaces used by drivers and administrators.
- Application Layer: Central Management Platform (CMP) responsible for processing business logic and sensor data.
- Data Layer: MySQL database used to store parking, payment, sensor, and user information.

3.0 SYSTEM ARCHITECTURE

3.1 Architectural Design

The Smart IoT-Based Parking System is designed using a modular three-tier architecture. The system is divided into several high-level subsystems to separate user interaction, business logic, sensor processing, and data storage.

The major subsystems are:

1. Driver Interface Subsystem
2. Administrator Interface Subsystem
3. IoT Sensor Subsystem
4. Central Management Platform (CMP)
5. Database Subsystem

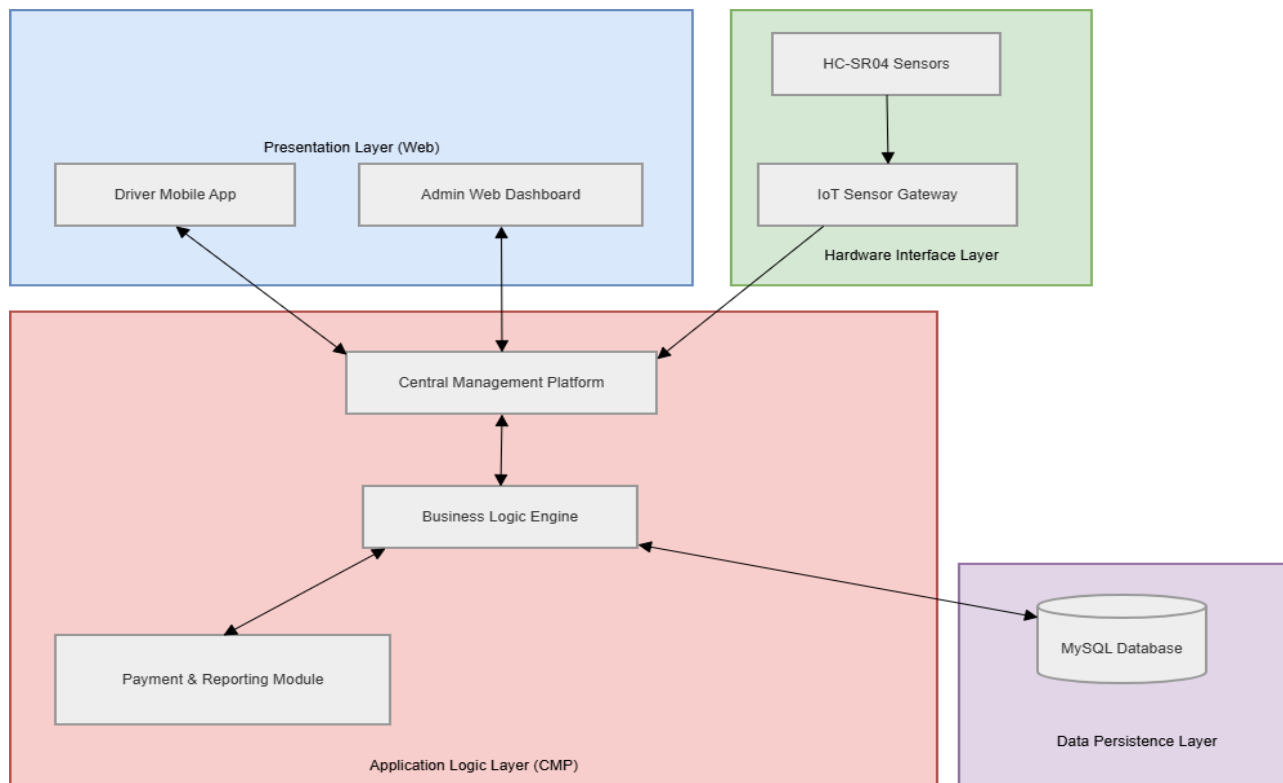


Figure 3.1

Figure 3.1 shows the three-tier architecture used in the Smart IoT-Based Parking System. The architecture consists of the Presentation Layer, Application Logic Layer, and Data Persistence Layer, which work together to provide real-time parking management services.

The Presentation Layer includes the Driver Mobile App and Admin Web Dashboard, where users interact with the system. Requests from users are sent to the Application Logic Layer through the API Gateway. The Application Logic Layer, also known as the Central Management Platform (CMP), processes user requests, manages parking availability, handles payments, and generates reports. At the same time, data from the HC-SR04 sensors is transmitted through the IoT Sensor Gateway to update parking slot status in real time.

Finally, the Data Persistence Layer contains the MySQL Database, which stores user information, parking slot records, sensor data, payment transactions, and reports. Through this architecture, all system components are able to communicate efficiently and provide accurate parking information to both drivers and administrators.

3.2 Decomposition Description

The Smart IoT-Based Parking System is decomposed into several major subsystems to ensure modularity, maintainability, and scalability. Each subsystem is responsible for a specific set of functionalities and interacts with other subsystems to provide complete system operations.

- i) **Driver Interface Subsystem**
 - This subsystem provides the user interface for drivers to interact with the system. It allows drivers to view real-time parking slot availability, access parking information, and interact with system services. It acts as the main communication layer between the driver and the system
- ii) **Administrator Interface Subsystem**
 - This subsystem is used by system administrators to monitor and manage the overall parking system. It provides functionalities such as managing parking areas, viewing system status, monitoring slot usage, and generating reports for analysis and decision-making.
- iii) **IoT Sensor Subsystem**
 - This subsystem is responsible for collecting real-time data from sensors installed at parking slots. It detects vehicle presence and continuously updates the status of each parking space (available or occupied). The collected data is transmitted to the Central Management Platform for processing.
- iv) **Central Management Platform (CMP)**
 - The Central Management Platform acts as the core processing unit of the system. It receives data from the IoT Sensor Subsystem, processes parking slot information, manages system logic, and distributes updated information to both driver and administrator interfaces.

v) Database Subsystem

- This subsystem is responsible for storing and managing all system data, including user information, parking slot status, transaction records, and system logs. It ensures data integrity, consistency, and provides reliable data access for other subsystems.

3.3 Design Rationale

The Smart IoT-Based Parking System uses the Centralized Client-Server Architecture to support the real-time parking management through an integrated platform. This architecture streamlined the mobile application of the driver's, admin web dashboard, Central Management Platform (CMP), IoT sensors and a centralized database into unified system. This model of architecture was chosen because it allows data synchronization between the parking availability information and the IoT sensors in real-time which will ensure all users have the access to a consistent parking information and payment data, On the other hand the admin able to manage operations through a centralized controller. This design help to simplify the system maintenance, optimization and future adjustment like additional parking spaces, zones and sensors can be interconnected with minor adjustment to the existing system.

4.0 Data Design

4.1 Data Description

Driver Interface Data: Stores the driver's information, active parking sessions, car plate number, and real-time payment transaction inputs for drivers.

Attribute	Type	Description
DriverID	Integer	Unique's driver identifier
Fullname	String	Driver's full name
Email	String	Driver's unique email address
CarPlateNo	String	Registered vehicle plate number
Password	String	Driver's login password
SessionID	Integer	Unique parking session identifier
StartTime	DateTime	Time when the vehicle enters the parking slot
EndTime	DateTime	Time when the vehicle leaves the parking slot.
PaymentID	String	Payment transaction identifier
Amount	Decimal	Total monetary fee computed or processed for the parking session.
PaymentMethod	String	Selected payment method
PaymentStatus	String	Current status of the transaction (e.g., Pending, Success, Failed).
PaymentDate	DateTime	Date and time when the payment transaction occurred.
ReceiptID	Integer	Unique identifier for the officially generated payment proof.

Subtotal	Decimal	Base parking cost before taxes.
TaxAmount	Decimal	Service tax and duties applied to the transaction.
TotalAmount	Decimal	Final billing amount charged to the user.
IssuedTime	Decimal	Timestamp record of when the receipt was created.

Administrator Interface Data: Stores the administrator's information, parking layout arrangement, slot status management, and argument for generating occupancy reports.

Attribute	Type	Description
AdminID	Integer	Unique's driver identifier
Fullname	String	Driver's full name
Password	String	Registered vehicle plate number
Email	String	Driver's login password
ReportID	String	Payment transaction identifier
Report_type	String	Parking fee amount
Amount	String	Selected payment method
SlotID	String	Unique alphanumeric identifier for a managed physical parking space.
Zone	String	Physical area, level, or sector location of the parking slot.
Slot_status	String	Real-time availability condition of the space.
Rate_per_hour	Decimal	Hourly pricing configuration assigned to the space.

IoT Sensor Data : Stores information about the IoT sensor that detects vehicle presence in each parking slot.

Attribute	Type	Description
SensorID	Integer	Unique sensor identifier
SlotID	Integer	Parking slot connected to the sensor
SensorStatus	String	Sensor condition such as Active, Inactive, or Error
VehicleDetected	Boolean	Shows whether a vehicle is detected or not
LastUpdated	DateTime	Latest time the sensor data was updated
LogID	Integer	Unique logging event sequence number.
StatusSent	String	Raw sensor data state payload pushed to the central system.
ErrorMessage	String	Hardware or transmission fault data text if a push fails.
LogTime	DateTime	The exact timestamp marking log file generation

Central Management Platform (CMP) Data: Manages the routes of data incoming, validation request received, calculated durations of parking, fee rates, system synchronization logs.

Attribute	Type	Description
SessionID	Integer	Active tracking session identifier matching vehicles to current coordinates.
PaymentStatus	String	State engine variable evaluating transaction validation responses.
SlotStatus	String	Available, Occupied, or Inactive
StatusSent	String	Parsing variable for validating received sensor data stream strings.
ErrorMessage	String	Hardware or transmission fault data text if a push fails.

Database Subsystem Data: Maintains the records of the driver's registered, administrative logs, payment records history, and long-term parking slot utilization history.

Attribute	Type	Description
DriverRecord	Record	Driver account information
PaymentRecord	Record	Payment transaction information
SlotRecord	Record	Parking slot information
SensorRecord	Record	Sensor information

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4.2 Data Dictionary

4.2.1 Admin Table

Data Element	Type	Size	Constraints	Description
AdminID	VARCHAR	30	Primary Key	Unique alphanumeric identifier for the administrator account within the system.
Admin_name	VARCHAR	100	Not Null	The full legal or display name of the authorized system administrator.
Password	VARCHAR	255	Not Null	The securely hashed authentication key required for the administrator to access the dashboard.
Email	VARCHAR	100	Not Null	The official contact email address used for system notifications, administrative communications, and account recovery.

4.2.2 Report Table

Data Element	Type	Size	Constraints	Description
ReportID	VARCHAR	30	Primary Key	Unique alphanumeric identifier for each system-generated report.
AdminID	VARCHAR	30	Foreign Key	References the AdminID in the ADMIN table, identifying the specific administrator who requested and generated the report.
Report_type	VARCHAR	50	Not Null	Specifies the designated timeframe or category of the generated report (e.g., 'Daily', 'Weekly',

				'Monthly').
Amount	DECIMAL	(10, 2)	Not Null	The total calculated financial revenue derived from parking transactions within the specified reporting period.

4.2.3 Parking Slot Table

Data Element	Type	Size	Constraints	Description
SlotID	VARCHAR	30	Primary Key	Unique alphanumeric identifier for each physical parking space managed by the system.
SensorID	VARCHAR	30	Foreign Key	References the specific IoT sensor assigned to monitor this space, linking the hardware to the database record.
Zone	VARCHAR	50	Not Null	The designated physical area, level, or sector where the parking slot is located (e.g., 'Level 1', 'Zone A').
Slot_status	VARCHAR	50	Not Null	The current real-time availability condition of the parking space (e.g., 'Available', 'Occupied', 'Inactive').
Rate_per_hour	DECIMAL	(5, 2)	Not Null	The specific pricing configuration or fee charged per hour for utilizing this parking slot.

4.2.4 Driver Table

Data Element	Type	Size	Constraints	Description
DriverID	INT	11	Primary Key	Unique driver identifier
Fullname	VARCHAR	100	Not Null	Driver's fullname
Email	VARCHAR	100	Unique, Not Null	Driver's email address

CarPlateNo	VARCHAR	20	Unique, Not Null	Registered vehicle plate number
Password	VARCHAR	255	Not Null	Encrypted password for login

4.2.5 ParkingSession Table

Data Element	Type	Size	Constraints	Description
SessionID	INT	11	Primary Key	Unique session identifier
CarPlateNo	VARCHAR	20	Foreign key	Registered vehicle plate number
SlotID	VARCHAR	30	Not Null	Unique alphanumeric identifier for each physical parking space managed by the system.
StartTime	DATETIME	-	Not Null	Time when the parking slot is entered by a vehicle
EndTime	DATETIME	-	Not Null	Time when the vehicle leave the parking slot.

4.2.6 Payment Table

Data Element	Type	Size	Constraints	Description
PaymentID	INT	11	Primary Key	Unique payment

				identifier
SessionID	INT	11	Foreign key,Not Null	Parking slot connected to sensor
CarPlateNo	VARCHAR	20	Not Null	Registered vehicle number
Amount	DECIMAL	10,2	Not Null	Detects vehicle presence
PaymentMethod	VARCHAR	50	Not Null	Selected payment method
PaymentStatus	VARCHAR	20	Not Null	Payment result
PaymentDate	DATETIME	-	Not Null	Date and time of the payment

4.2.7 Receipt Table

Data Element	Type	Size	Constraints	Description
ReceiptID	INT	11	Primary Key	Unique Identifier for generated driver's official payment receipt
PaymentID	INT	11	Foreign Key, Not Null	References PaymentID in the PAYMENT table to map the invoice directly to a verified transaction
DriverID	INT	11	Foreign Key, Not Null	References DriverID in the Driver table to link the bill to the specific car owner's profile.
Subtotal	DECIMAL	5,2	Not Null	The base cost calculated from the hours parked multiplied by the slot hourly rate.
TaxAmount	DECIMAL	10,2	Not Null	Service tax

				duties applied automatically to the transaction
TotalAmount	DECIMAL	10,2	Not Null	Service tax or municipal parking duties applied automatically to the transaction.
IssuedTime	DECIMAL	-	Not Null	The final gross payment amount charged to the user,

4.2.8 IoT Sensor Table

Data Element	Type	Size	Constraints	Description
SensorID	INT	11	Primary Key	Unique sensor identifier
SlotID	INT	11	Foreign key,Not Null	Parking slot connected to sensor
SensorStatus	VARCHAR	20	Not Null	Current sensor condition
VehicleDetected	BOOLEAN	-	Not Null	Detects vehicle presence
LastUpdated	DATETIME	-	Not Null	Latest sensor update time

4.2.9 Parking_Status Table

Data Element	Type	Size	Constraints	Description
SlotID	INT	11	Primary key	Unique parking slot
SensorID	INT	11	Foreign Key	Sensor monitoring the slot

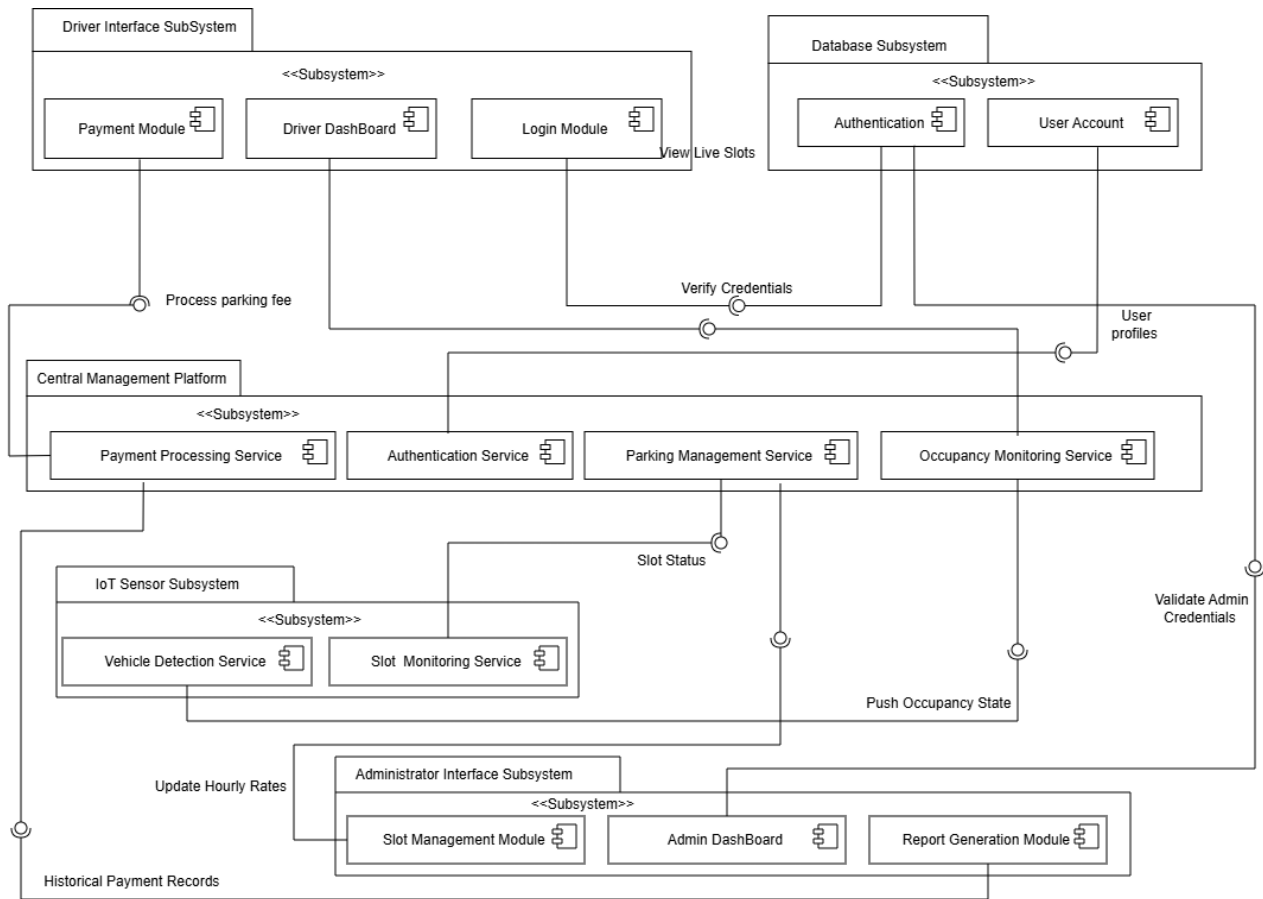
Zone	VARCHAR	50	Not Null	Parking area or zone
SlotStatus	VARCHAR	20	Not Null	Available, Occupied, or Inactive
LastUpdated	DATETIME	-	Not Null	Latest slot status update

4.2.10 Sensor Log

Data Element	Type	Size	Constraints	Description
LogID	INT	11	Primary key	Unique log identifier
SensorID	INT	11	Foreign Key, Not Null	Related IoT sensor
StatusSent	VARCHAR	20	Not Null	Sensor status sent to system
ErrorMessage	VARCHAR	255	Null	Error message if data transmission fails
LogTime	DATETIME	-	Not Null	Time the log is created

5.0 Component Design

Component Design for Smart IoT-Based Parking System -> developed by interconnected software and hardware components.



5.1 Driver Interface Subsystem

Responsibilities:

- Allow drivers to log in using car plate number and password.
- Display real-time parking availability.
- Process parking payments.
- Show driver dashboard with active session.

Inputs:

- Car plate number and password
- Parking slot selection
- Payment method

Outputs:

- Login status
- Available parking slots
- Payment receipt
- Driver session information

Pseudocode:

START

Enters car plate number and password

IF credentials are valid THEN

 Display driver dashboard

 Show active parking session (if any)

ELSE

 Display login error message

ENDIF

IF driver selects "View Slots" THEN

 Retrieve and display available parking slots

ENDIF

IF driver selects "Make Payment" THEN

 Calculate parking fee

 Process payment

 Generate receipt

ENDIF

END

5.2 Administrator Interface Subsystem

Responsibilities:

- Monitor parking occupancy
- Manage parking slots
- Generate reports

Inputs:

- Admin login information
- Slot information
- Report type and date range

Outputs:

- Occupancy dashboard
- Updated records of slot
- Generated reports

Pseudocode:

START

Logs into system

Enter credentials needed

IF admin credentials are valid THEN

 Display admin dashboard

```
IF "Monitor Occupancy" is selected THEN
    Display total slots, occupied, and available
    Show heat map visualization
ENDIF

IF "Manage Slots" is selected THEN
    Add new slot OR Update existing slot OR Delete slot
    Save changes to database
ENDIF

IF "Generate Report" is selected THEN
    Select report type and date range
    Retrieve records from database
    Display and download report
ENDIF

ELSE
    Display login error message
ENDIF

END
```

5.3 IoT Sensor Subsystem

Responsibilities:

- Detect vehicle availability in the parking slots
- Update parking status in real-time

- Send data to the Central Management Platform(CMP)

Inputs:

- Sensor readings
- Data of vehicle detection

Outputs:

- Parking slot status
- Real-time data transmission

Pseudocode:

```
START

Detects vehicle presence

IF vehicle is detected THEN
    Set status = "OCCUPIED"
ELSE
    Set status = "AVAILABLE"
ENDIF

Send status to Central Management Platform

Update database with new status

IF transmission fails THEN
    Retry 3 times
    IF still fails THEN
```

```
        Mark slot as "INACTIVE"

        Log error for maintenance
    ENDIF
ENDIF
END
```

5.4 Central Management Platform (CMP)

Responsibilities:

- Validate user's credentials
- Parse parking data from sensors
- Calculate Parking fees
- Process Payments
- Manage parking operations

Inputs:

- Login credentials
- Sensor data
- Payment requests

Outputs:

- Authentication status
- Updated parking availability
- Payment confirmation

Pseudocode:

```
START
```

```
IF login request received THEN
    Validate credentials against database
    Return authentication status
ENDIF

IF sensor data received THEN
    Process parking status
    Update database
    Push update to driver interface
ENDIF

IF payment request received THEN
    Retrieve parking session
    Calculate fee based on duration and rate
    Process payment through gateway
    Save transaction record
    Generate receipt
ENDIF

END
```

5.5 Database Subsystem

Responsibilities:

- Store and manage all the data in the system

- Provide access of data for the other subsystems
- Ensure data integrity and consistency

Inputs:

- Data queries
- Insert/ Update/ Delete requests

Outputs:

- Retrieved data
- Operation Status

Pseudocode:

```

START

IF query is received THEN
    Retrieve data from appropriate table
    RETURN result
ENDIF

IF insert request is received THEN
    Validate data
    Insert into database
    RETURN success status
ENDIF

IF update request is received THEN
    Validate data

```

```

Update database record
RETURN success status
ENDIF

IF delete request is received THEN

    Confirm deletion

    Delete record from database

    RETURN success status

ENDIF

END

```

6.0 HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

6.2 Screen Images

I) Driver Interface

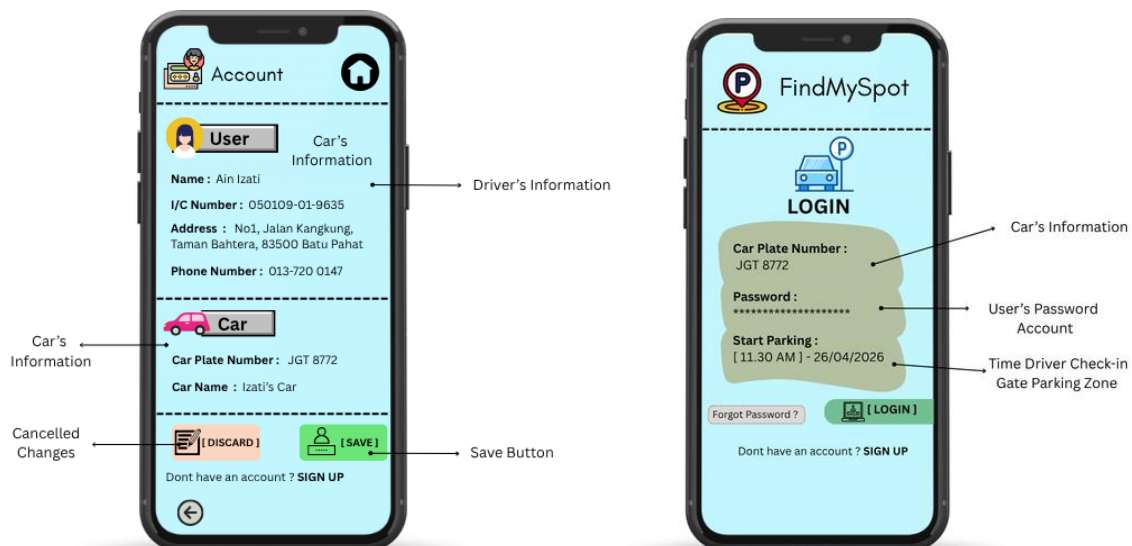


Figure 6.2.1 Account Management

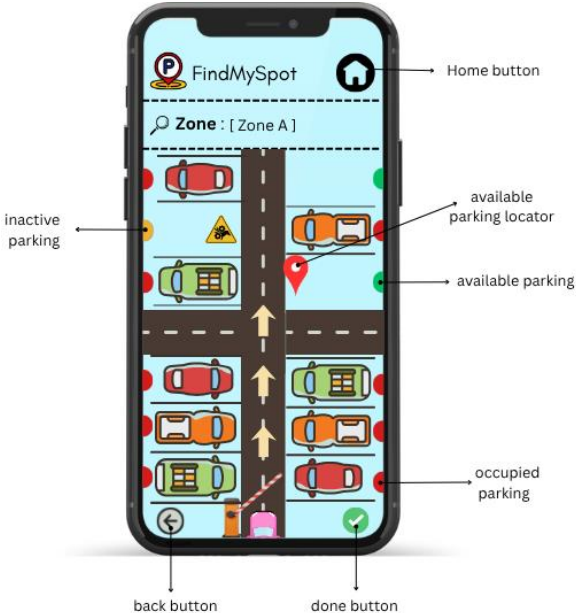


Figure 6.2.3 Real-Time Parking

Figure 6.2.2 Login Account



Figure 6.2.4 Payment

II) Admin Interface

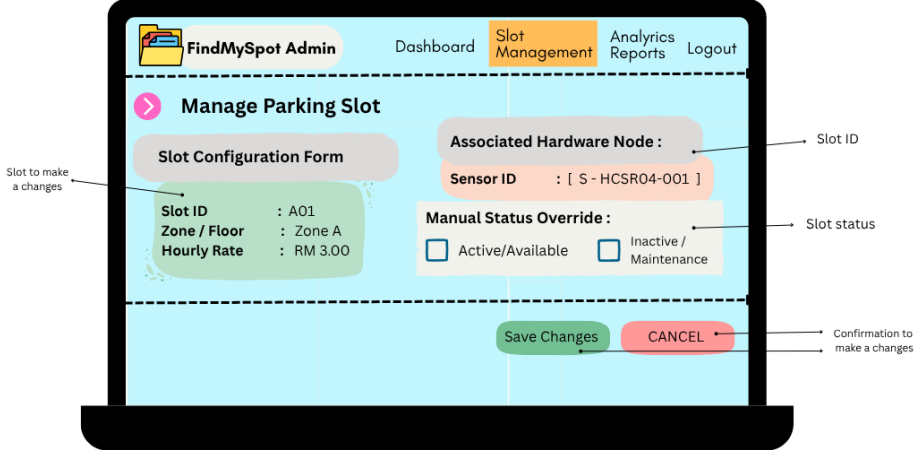


Figure 6.2.5 : Manage Parking Slot

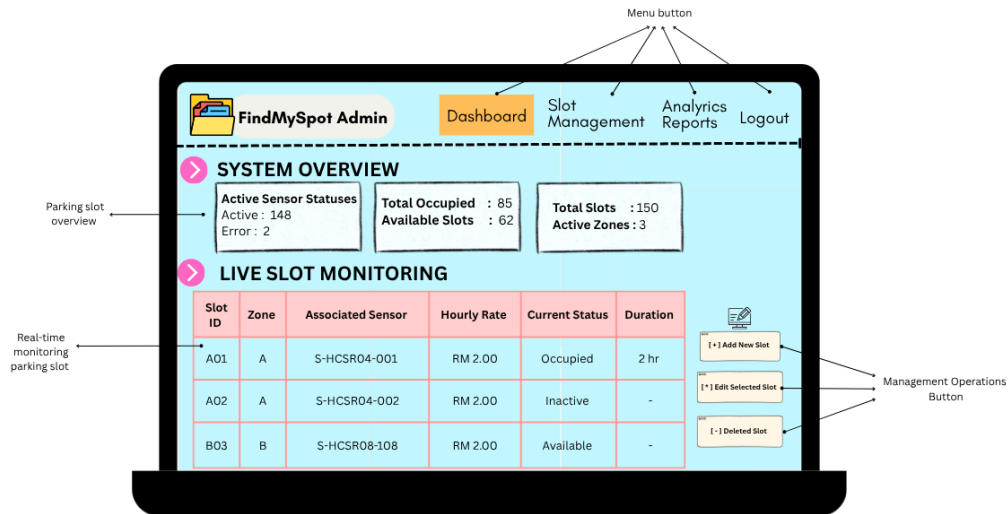


Figure 6.2.6 Admin Dashboard

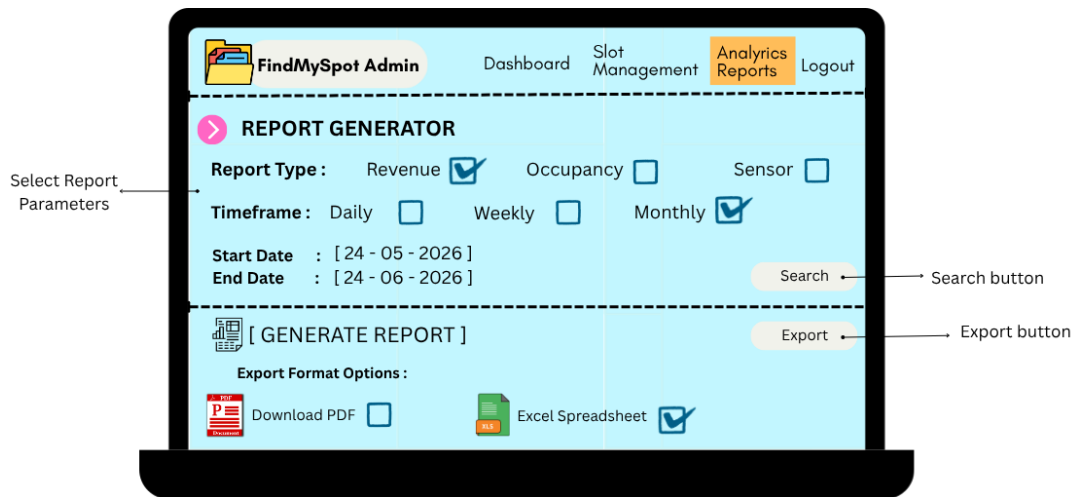


Figure 6.2.7 Generated Report

6.3 Screen Objects and Actions

7.0 REQUIREMENTS MATRIX

This section provides a cross-reference matrix that traces the system components and database structures designed in this document back to the functional requirements specified in the Software Requirements Specification (SRS) document. This ensures that every functional requirement defined in the SRS has been properly addressed by at least one system component and data structure, verifying the completeness and accuracy of the system design.

SRS Requirement ID	Requirement Description / Feature	System Component(s) Satisfying	Associated Data Structure(s)
--------------------	-----------------------------------	--------------------------------	------------------------------

		Requirement	
REQ-1.1 to REQ-1.4	User Login Authentication: Allow drivers and administrators to securely log in, validate credentials, and route to the correct dashboard.	• Driver Interface Subsystem • Administrator Interface Subsystem • CMP: Authentication Service • Database Subsystem	• Driver Table • Admin Table
REQ-2.1 to REQ-2.6	View Parking Availability: Request sensor data and display highlighted real-time parking slot vacancies or "Full" notifications.	• Driver Interface Subsystem • CMP: Parking Management Service • Database Subsystem	• Parking_Slot Table • Parking_Status Table
REQ-3.1 to REQ-3.7	Make and View Payment: Calculate fees, process transactions, generate receipts, and display historical payment records.	• Driver Interface Subsystem • CMP: Payment Processing Service • External Payment Gateway • Database Subsystem	• Payment Table • Receipt Table • ParkingSession Table
REQ-4.1 to REQ-4.3	Monitor Parking Occupancy: Display real-time occupancy, heat map visualizations, and trigger maintenance alerts for failed sensors.	• Administrator Interface Subsystem • CMP: Occupancy Monitoring Service • Database Subsystem	• Parking_Slot Table • Parking_Status Table • IoT Sensor Table
REQ-5.1 to REQ-5.6	Manage Parking Slots: Add, update, delete, and validate parking slots while rejecting duplicate slot	• Administrator Interface Subsystem • Admin: Slot Management Module	• Parking_Slot Table

	IDs.	• Database Subsystem	
REQ-6.1 to REQ-6.6	Generate Reports: Retrieve historical records, compile daily/weekly/monthly revenue reports, and export to PDF/Excel.	• Administrator Interface Subsystem • Admin: Report Generation Module • Database Subsystem	• Report Table • Payment Table • Parking_Slot Table
REQ-7.1 to REQ-7.7	Update Parking Availability: Automatically process incoming sensor data, refresh availability, and log transmission errors.	• IoT Sensor Subsystem • CMP: Parking Management Service • Database Subsystem	• Parking_Status Table • IoT Sensor Table • Sensor Log Table
REQ-8.1 to REQ-8.6	Send Real-Time Data: Detect vehicle presence physically, determine status, and transmit continuous data to the CMP over IoT network.	• IoT Sensor Subsystem (Vehicle Detection Service & Slot Monitoring Service) • CMP: Parking Management Service	• IoT Sensor Table • Sensor Log Table

8.0 APPENDICES

This section is optional.

Appendices may be included, either directly or by reference, to provide supporting details that could aid in the understanding of the Software Design Document.